

Web XR User Interface Study in Designing 3D Layout Framework in Static Websites

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ABSTRACT

With computer technology rapidly expanding out in recent years, there is a significant trend that Virtual Reality, Augmented Reality, and Mixed Reality technologies turn into the public. The paper describes Web Extended Reality (XR) and its current circumstance. The research discussed the advantages of web componentization and Page Builder System, which is the famous framework for web componentization. Furthermore, the research analyzes the characteristic of XR. The research designs the Web XR User Interface principles with XR characteristics and componentization design. The principles cover three aspects include main content, scrollbar, and navigation. The research develops the prototype to examine the concept. The prototype shows that the UI principles can provide an immersive user experience. The paper indicates the possible future view based on the design study.

Keywords: Virtual Reality, User Interface, Static Websites, Web.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality, Mixed /augmented reality

1 INTRODUCTION

The Immersive Web Working Group under the W3C has announced the Web XR Device Application Programming Interface (API) specifications in 2018 [1 2]. Web XR is based on HTML5 technology and simulates a three-dimensional (3D) rendering of the virtual world in the browser [3]. Web XR Device API allows the web application to access the XR device motion and direction tracking features as the native applications [1][3].

Compared to mobile phones, XR devices are more comfortable to carry and closer to natural interaction than mobile phones. However, Web XR still has a particular gap in achieving the full panoramic 3D display and interactive effects due to the platform limitation in browsers and devices. Consequently, XR devices have limited support to transfer websites into XR platforms. For instance, HoloLens supports controlling the Edge browser from the remote computer [4]. Edge supports basic HoloLens gestures

include scroll, drag, and zoom but the UI visual effect is the same as the desktop version [4]. Significantly, the desktop version is not adapted to XR platforms because their interactions are designed from keyboard/mouse, not controller/gesture. Using the controller/gesture to complete these interactions is inconvenient and brings negative user experiences.

Considering the above evidence, although Web XR has a broader prospect, it still has limitations. It sits in the initial developing period [5]. The limitation brings negative user experiences and extends the user group in the Web XR. Therefore, designing an efficient and immersive interactive framework in Web XR is necessary to contribute to the Web XR field.

There are various types of websites globally. It is necessary to narrow the research scope into a specific category and analyze its related UI structure and design framework. The various websites can divide into two types: dynamic and static [6]. Web Hosting statistics and facts for 2021 that the static websites without e-Commerce functions occupy 51.3% [7]. However, WPFForms, one of the largest CMS plugin providers globally, statistics that medium/small businesses, the owners of the static websites, usually cannot afford a high investment in online business [8]. Therefore, it is challenging to drive them to jump into the Web XR field and improve their online service because XR still has high barriers for development. The research focuses on static websites (HTML5) to build the initial design principle with fundamental XR characteristics. The research tries to provide a general and affordable UI framework to assist the static websites to transfer into the Web XR field.

2 FRAMEWORK OVERVIEW

In this project, the research aims to design a general Web XR UI framework. It can apply to small/medium businesses and organizations' websites to improve their user experiences in XR platforms. The proposed principle is based on XR interaction methodologies. The characteristics are as follows:

1. The research uses the HTML5 and XR interaction features to build the UI principle and provide users with immersive exploratory experiences.
2. The research balances the website's original UI structure and XR characteristics. The research uses responsive design to display the same content as desktop websites. Meanwhile, 3D layout and gesture navigation are integrated with pages to enhance the user's immersive experience.
3. The research evaluates the performance efficiency of the system. We conducted performance evaluations and a user study.

The session describes how to build up the web UI principles and investigate the related XR UI knowledge in this section. Furthermore, the research designs a general 3D layout principle in the main content. At first, the layout aims to fit with various

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HTML5 static websites. Secondly, the research analyzes how to use gesture interaction and 3D space to improve the web navigation system.

The research uses Unreal 4's MR UX development tools to design interactive website prototypes based on XR devices. The research uses GNA Web Design Websites to testify the possibility of the principles. The prototype can display how the concept plan adapts to the XR environment and the existing website.

2.1 Design Concept

Static Websites mainly assist the visitor to find helpful information efficiently. Therefore, the website has a general system structure. It usually contains Home, About us, Contact and other information pages [9]. Each information page has a similar layout structure. The general structure covers main content and navigation [10]. Most of the existing web design principles are based on building up the general layout to adapt to different business criteria. The research focuses on building the 3D layout for the main content and navigation.

Responsive layout is a concept proposed by Ethan Marcotte in 2010 [11]. The primary concept is that an HTML5 website can be compatible with multiple digital devices rather than making a specific version for each terminal. This concept was born to solve the problem of mobile Internet browsing. Responsive layout can provide a more comfortable interface and better user experience in various devices. Therefore, the research can use the responsive layout to customize the XR UI principle in Web XR platforms.

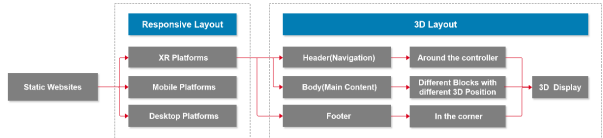


Figure 1: Static Websites 3D Layout Concept.

Considering the above static websites background, the research designs the 3D layout solution concept (Figure 1) to apply the XR characteristic. The static websites (HTML5) use responsive layout to divide different platforms' visual effects. The navigation displays in front of the user while activating the menu button. The navigation information displays a 3D perspective to guide the user to explore the websites. The Body section (main content) divides the content into different blocks. Different blocks display in different positions in the 3D environment. The footer displays in the bottom corner of the 3D environment because it is not the critical but essential element to the websites. The concept provides a 3D visual effect in highlighting navigation and main content.

2.2 Main Content

The 3D layout requires dividing the main content of the page and avoid the user's distraction. We refer to the Page Builder system to divide the page into different parts. Each part can be presented to users in different positions and angles as independent modules to achieve a perspective visual effect. After dividing the modules, we need to place each module in 3D environments.

Figure 2 displays the main content structure concept. The main content divides into three sessions: current visiting block, other blocks and scrolling bar. The current block positions in the middle of the screen while other blocks stay at up and down corners with specific angles. It uses transparency to weaken other blocks and avoid distracting users' attention when visiting the current block. The scroll bar is on the right side with touching interaction.

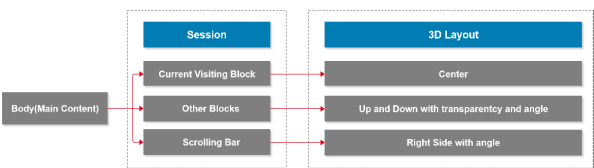
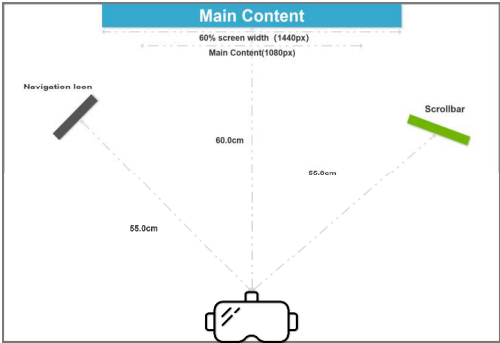
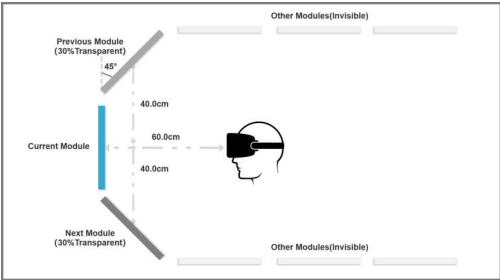


Figure 2: Main Content Structure Concept.

Figure 3(a) shows that the current module is 60.0 cm far from users and follows the user movement. It does not follow the user's head rotation so that the users can look around the UI. The scroll bar and Navigation icon are 55.0 cm distance from users. The previous and next modules have a 45-degree rotation and 30% transparency (Figure 3(b)). The user can use scrolling interaction to move different modules (Figure 3(c)).



(a)



(b)

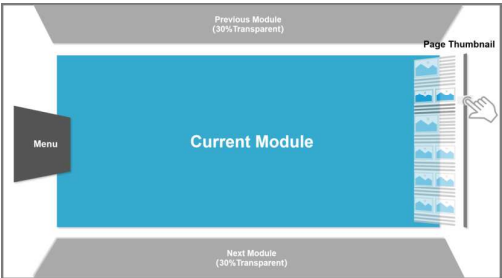


(c)

Figure 3: Main Content Layout Plan: (a) Top View (Concept); (b) Side View (Concept); (c) Developing Screenshot (Scrolling the Page).

Figure 4(a) displays the XR scrollbar concept solution. When the user's hand controller drags the scrollbar or the main content, a thumbnail of the web page will display on the right side of the scrollbar (Figure 3(c) and Figure 4(a)). The current module is highlighted. Users can briefly understand the page content and

current location through the page thumbnails. The scrollbar thumbnails only exist when the user scrolls the slider. This solution does not disturb the user's focus and provides clear guidance.



(a)



(b)

Figure 4: Scrollbar Design: (a) Scrolling Concept; (b) Developing Screenshot (Scrolling the bar).

2.3 Navigation

The traditional web menu provides users with basic button options without detailed location information. However, the menu should have the navigation's feature that assists the user to understand “where he is”. Therefore, our research aims to use 3D layout features to provide clear guidance for users. Users can use gesture interaction to select the link conveniently.

Figure 5 displays the navigation interaction flow concept. The detailed navigation information is invisible in the initial loading. The navigation icon is on the left side in Figure 3(a). When the user clicks the icon, the location panel and a 3D ring will display. By sliding gestures and rotating the 3D ring, the user can switch between location and menu panel. Besides, each location panel provides “Home” button to jump to the menu panel. Both pages provide links to jump to the web pages.

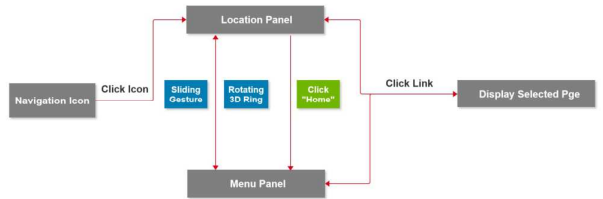
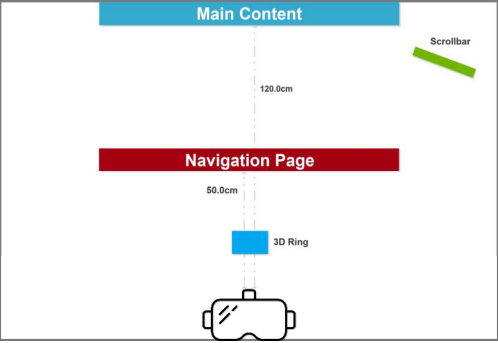


Figure 5: Navigation Interaction Flow Concept.

Figure 6 illustrates how the navigation fits into the 3D layout and gesture. When the user clicks the menu icon, the main content and scrollbar move far away and the navigation panel will display in front of the user as shown in Figure 6(a). The user can rotate the 3D ring or sliding gesture to switch the position and menu panel (Figure 6(b)).



(a)



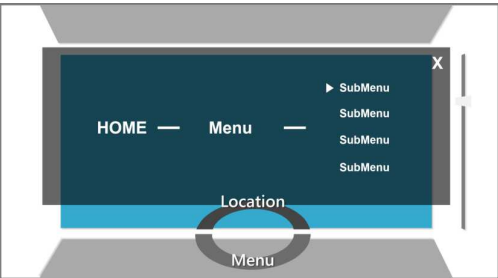
(b)

Figure 6: Navigation Design: (a) Top View in Navigation Position; (b) How to Switch Between Location and Menu Panel.

Figure 7 displays the navigation panel details. The location panel displays the current location and provides other links in the same sublevel (Figure 7(a)& Figure 7(b)). The page will jump to the selected page when the user selects other submenus. The location panel can guide the user to understand their location and assist them fast in selecting the related page.

At the same time, the framework provides enough convenience to help the user move to other menus. By sliding or rotating the 3D ring, the user can switch to the menu panel and check other levels' pages. Furthermore, the user clicks the 'Home' button will also switch to the menu panel and check the menu list (Figure 7 (d)).

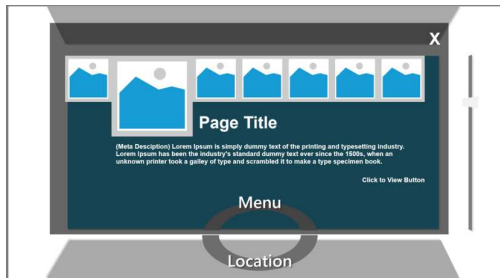
The menu panel provides detailed meta descriptions to assist the user to know each submenu page's key information (Figure 7 (c)). Users can check each menu detail by clicking different menu thumbnails.



(a)



(b)



(c)



(d)

Figure 7: Navigation Details: (a) Location Panel Concept; (b) Location Panel Prototype; (c) Menu Panel Concept; (d) Menu Panel Prototype.

3 CONCLUSION

This paper describes Web XR technologies. It illustrates the current circumstance for web browser services in the XR platform. The study designs the affordable and responsive framework concept for the static website in XR platforms. The UI principles follow the web componentization rules to arrange the 3D layout. The UI principles cover main content, scrollbar, and menu. The study developed the prototype to examine the theoretical principle. The prototype shows that the UI principles can provide an immersive user experience.

4 FUTURE WORK

The research is still under processing. Although we have built up the Web XR prototype, we have not recruited large group users to start the formal experiment in VR and MR platforms. Furthermore, the UI principles require further user feedback to improve its interactions.

In the future, we will continue to improve the framework. We will run a large group of user tests to receive the feedback to optimize the interactive functions through user feedback. We hope that the research can contribute to the Web XR field.

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